

Chapter 6

Optimal control problem applied for wave energy converters

6.1 Modelling of a hemispherical heaving floater

The aim of this section is to implement a python code capable of representing the dynamics of a hemispherical heaving floater Wave Energy converter coupled to a linear Power take off, as show in Fig. 6.1.

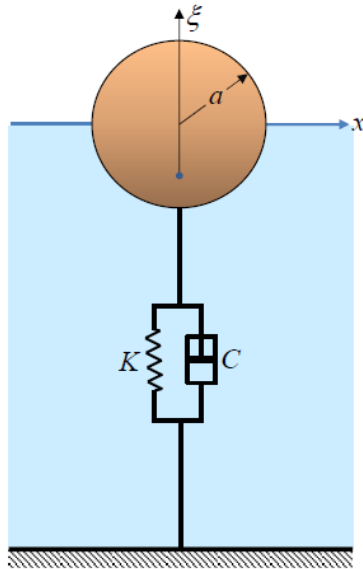


Figure 6.1: Hemispherical heaving floater [1]

The dynamics equation of this system is usually described by the so called Cummins equation for wave energy absorption by oscillating bodies:

$$[m + A(\infty)]\ddot{\xi}(t) = f_e(t) - \int_{-\infty}^t g_r(t - \tau)\dot{\xi}(\tau) d\tau - \rho g S_{cs}\xi(t) + f_{PTO} \quad (6.1)$$

The initial assumptions of this simplified approach involve first solving the equations of dynamics in the time domain. The convolution integral will be approximated with a constant equal to the damping coefficient $B(\omega)$. As far as the added mass is concerned, the value relative to the infinite frequency $A(\infty)$ will be considered.

The dynamics of the buoy is thus described by the following equation

$$(m + A(\infty))\ddot{\xi}(t) + (B(\omega) + C)\dot{\xi}(t) + (\rho g S_{cs} + K)\xi(t) = f_e(t) \quad (6.2)$$

where

$$f_e(t) = \Gamma A_w \cos(\omega t) \quad (6.3)$$

is the wave excitation force with a certain wave height A_w and a wave frequency $\omega = 2\pi/T$, given a wave period T .

In resonance conditions:

$$\begin{cases} C = B(\omega) \\ \omega = \sqrt{\frac{\rho g S_{cs} + K}{m + A(\infty)}} \end{cases} \quad (6.4)$$

so we can obtain the PTO stiffness coefficient as:

$$K = \omega^2(m + A(\infty) - \rho g S_{cs}) \quad (6.5)$$

It is worth to note that a negative value of K could be required in order to assess the resonance frequency condition. Choosing C and K values that differ from the above calculations should take us not to resonance conditions. Then, maximum power will not be achieved.

Fig. 6.2 and Fig. 6.3 show the displacement and velocity trend for a hemispherical buoy of radius $a = 5$ m subjected to a wave of $T = 9$ s and $A_w = 1$ m, not in resonance and in resonance conditions respectively. As can be seen, choosing the right values of C and K (as in Eq. (6.4)) leads to resonance conditions and thus to achieving the maximum power that can be extracted from the system.

It is worth to make a comparison between the time domain analysis and the frequency domain analysis. In the frequency domain, the buoy dynamics equation becomes:

$$\{-\omega^2(m + A(\omega)) + i\omega(B(\omega) + C) + (\rho g S_{cs} + K)\} X = F_e \quad (6.6)$$

In practice, in the approach used, a difference between the Eq. (6.2) and the Eq. (6.6) lies in the choice of the added mass value, as in the frequency domain it is calculated as a function of the wave frequency ω .

It is expected to obtain minimal differences between one analysis and the other, since in any case the approximation made on the convolution integral is acceptable in the first instance. In fact, a comparison between the time domain and the frequency domain analysis is shown in Fig. 6.4 and in Fig. 6.5, not in resonance and in resonance conditions respectively.

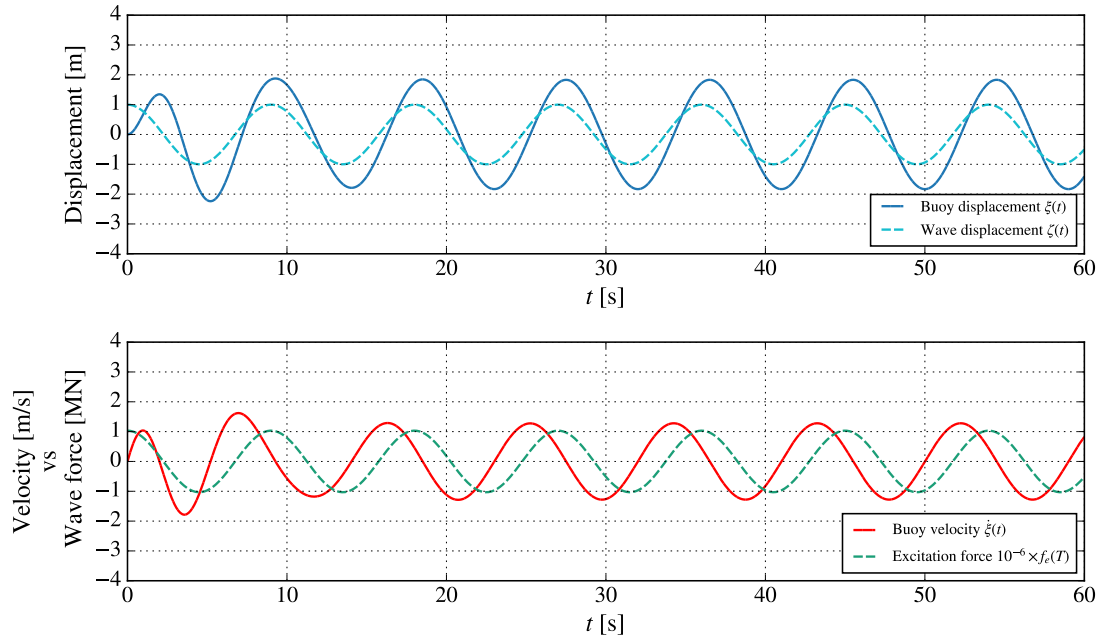


Figure 6.2: Results for a hemispherical buoy of radius $a = 5$ m subjected to a wave of $T = 9$ s and $A_w = 1$ m, not in resonance conditions

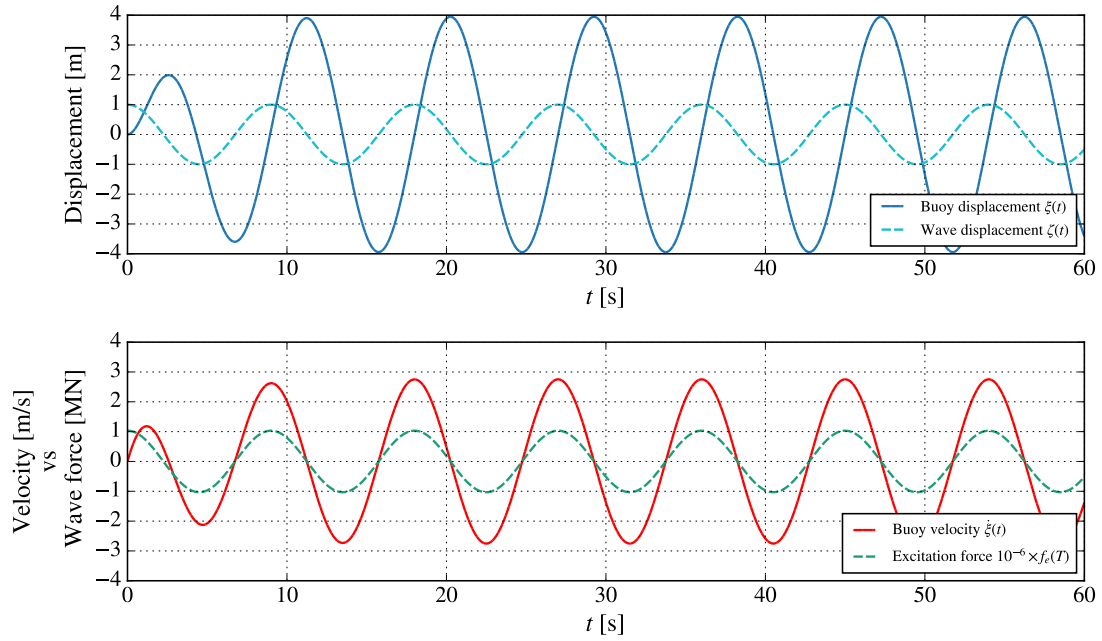


Figure 6.3: Results for a hemispherical buoy of radius $a = 5$ m subjected to a wave of $T = 9$ s and $A_w = 1$ m, in resonance conditions

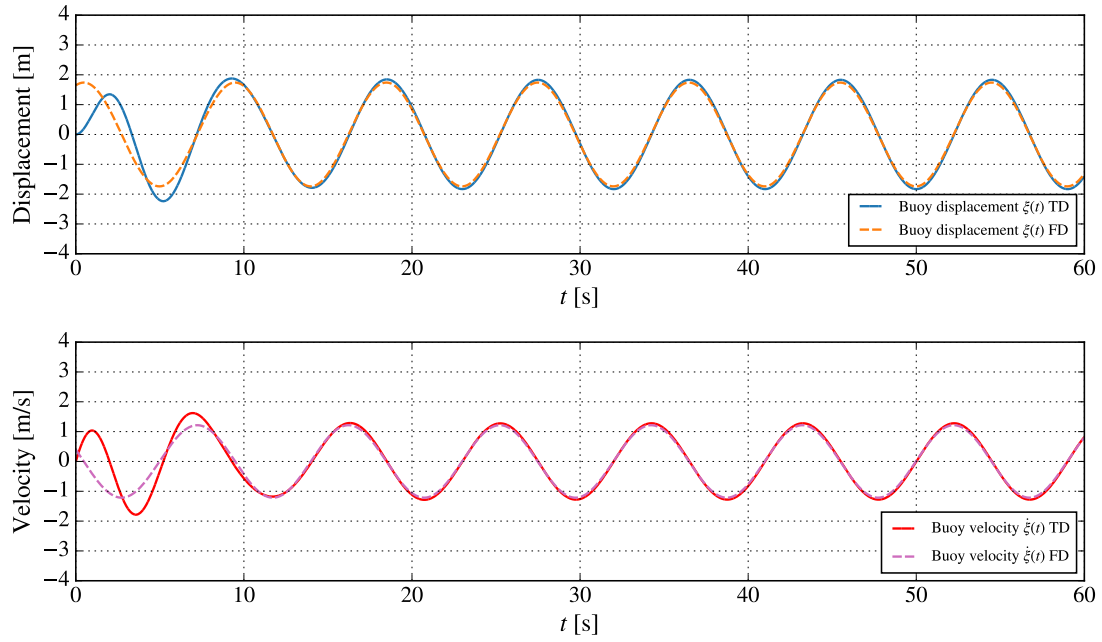


Figure 6.4: Comparison of time domain and frequency domain for buoy displacement and speed, not in resonance conditions

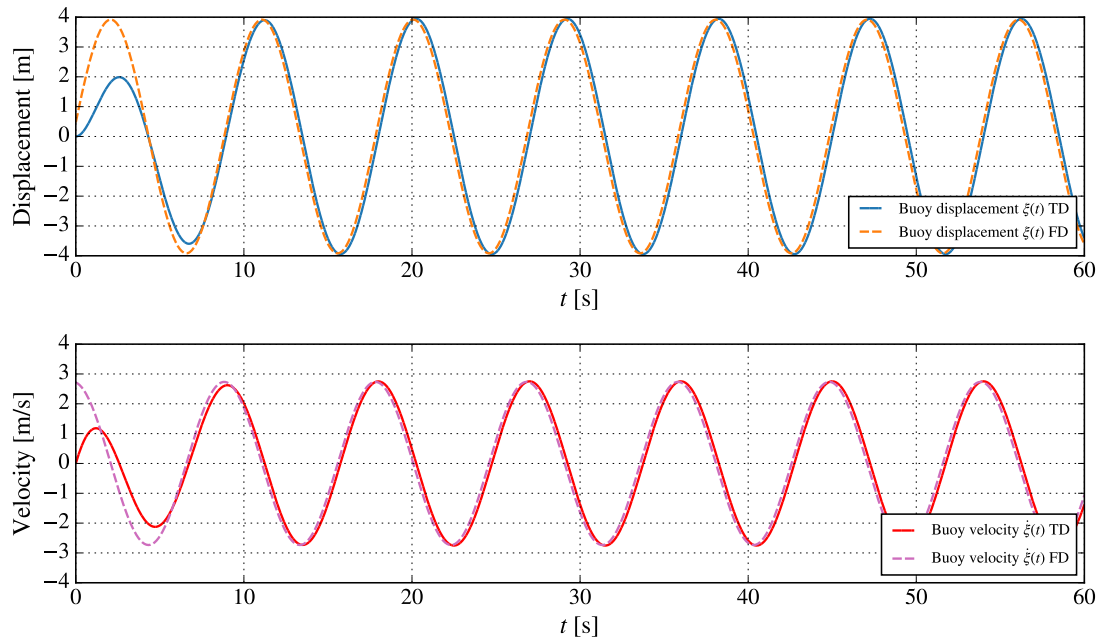


Figure 6.5: Comparison of time domain and frequency domain for buoy displacement and speed, in resonance conditions

6.2 Latching Control

As it is well known, in the early 1980s, Budal and Falnes [8] demonstrated that, for point absorbers, one key condition for maximizing energy production is maintaining the device's velocity in phase with the excitation force. To achieve this synchronization, they proposed a phase-control strategy known as *latching control*, specifically applied to their heaving buoy wave energy converter. Latching control involves temporarily immobilizing the motion of the buoy precisely at the instant its velocity reaches zero at the end of an oscillation cycle, subsequently releasing it at the optimal moment. Determining the duration for which the buoy should remain locked represents the central challenge to be addressed.

When latching control is implemented, an additional force must be introduced into the dynamics of the device (Eq. (6.2)) to nullify the acceleration of the controlled motion. Furthermore, since latching control is activated precisely when the velocity becomes zero, the equation of motion during the locking phase is expressed as follows:

$$\begin{cases} \dot{\xi}(t) = 0 \\ \ddot{\xi}(t) = 0 \end{cases} \quad (6.7)$$

6.3 Optimal latching control for a hemispherical heaving floater

It is possible to adopt an optimal control theory based on Pontryagin's maximum principle, as introduced in this context by Hoskind and Nichols [9], for latching control strategy.

To efficiently implement this theory numerically, latching control, which inherently represents an absolute bang-bang strategy, must be approximated using a less stringent model. In the ideal absolute latching scenario, a force is applied to the device that instantaneously nullifies its acceleration. However, in the weaker formulation, this instantaneous force is substituted by a very large, yet finite, damping force. Thus, latching control is modeled by introducing a substantial control force into the equation of motion:

$$f_{latching}(t) = -Gu(t)x_j(t) \quad (6.8)$$

where x_j denotes the j -th component of the state vector $\mathbf{x}$, representing the velocity of the controlled motion, G is the large damping coefficient, and u is the control variable. When $u = 0$, latching control is inactive; conversely, when $u = 1$, it is activated.

Given a sequence of wave excitations over an extended period $[0, T]$, the objective is to identify the optimal binary control sequence $u(t)$ that maximizes energy production:

$$\max_u E(u) = \int_0^T Q(t, \mathbf{x}, u) dt \quad (6.9)$$

where $Q(t, \mathbf{x}, u)$ is the instantaneous absorbed power.

6.3.1 First problem formulation attempt

First, from the dynamics equation in time domain Eq. (6.2), we must consider the latching control term:

$$(m + A(\infty))\ddot{\xi}(t) + (B(\omega) + C)\dot{\xi}(t) + (\rho g S_{cs} + K)\xi(t) = f_e(t) + f_{latching}(t) \quad (6.10)$$

Let $z(t) = \xi(t)$ and $v(t) = \dot{\xi}(t)$, then the state vector $\mathbf{x}$ can be written as:

$$\mathbf{x} = \begin{pmatrix} z \\ v \end{pmatrix} \quad (6.11)$$

Hence, if we also wish to lighten the notation by subtending the time dependency t , Eq. (6.10) becomes:

$$(m + A(\infty))v + (B(\omega) + C + Gu)v + (\rho g S_{cs} + K)z = f_e \quad (6.12)$$

being $f_{latching} = -Guv$ and:

$$G = G^*(m + A(\infty)) \quad (6.13)$$

with $G^* \in \mathbb{R}^+$, measured in s^{-1} and chosen appropriately large. Moreover, the equation of motion (6.12) can be written under the state form (2.1):

$$\begin{cases} \dot{z} = v \\ \dot{v} = F_v(t, u, v, z) \end{cases} \quad (6.14)$$

where the right-hand side of the second equation in (6.14) has been written as:

$$F_v(t, u, v, z) = \frac{f_e - (B(\omega) + C + Gu)v - k_{tot}z}{m_{tot}} \quad (6.15)$$

being $m_{tot} = m + A(\omega)$ and $k_{tot} = \rho g S_{cs} + k$.

Finally, the instantaneous absorbed power can be calculated as:

$$Q(t, \mathbf{x}, u) = f_{PTO}v = Cv^2 \quad (6.16)$$

Thus, the goal of this optimization will be the following:

$$\max_u E(u) = \int_0^T Cv^2 dt \quad (6.17)$$

We can now define the Hamiltonian according to Eq. (2.10):

$$H = Cv^2 + \lambda_1 v + \lambda_2 F_v(t, u, v, z) \quad (6.18)$$

It is useful to note that the Hamiltonian function represents a power, so it is measured in W. Furthermore, the first Lagrange multiplier λ_1 is measured in N, representing a force, while the second one λ_2 in kg·m/s, representing a momentum.

From Eq. (2.19), the adjoint state equations for this problem are given by:

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial z} = -\lambda_2 \frac{k_{tot}}{m_{tot}} \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial v} = -2Cv - \lambda_1 + \lambda_2 \frac{(B(\omega)+C+Gu)}{m_{tot}} \end{cases} \quad (6.19)$$

Since the Hamiltonian is a linear function of the control variable u , the resulting optimal control is of the "bang-bang" type. This implies that, in order to maximize the Hamiltonian H , the control variable u must adopt one of its extreme values, either 0 or 1. Examining Eq. (6.18), it follows that at any given instant, the Hamiltonian reaches its maximum under the condition:

$$\frac{\partial H}{\partial u} = 0 \iff -\lambda_2 Gv = 0 \quad (6.20)$$

hence:

$$u = \begin{cases} 1 & \text{if } -\lambda_2 Gv > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (6.21)$$

By employing Eqs. (6.14), (6.19), and (6.21), it is possible to derive the optimal control sequence $u(t)$ for any type of incident wave—whether regular or irregular—over the time interval $[0, T]$.